

12th European Conference on Service-Oriented and Cloud Computing

Analytics Feature Space: a Novel Framework for Interoperable Edge Machine Learning Detection

Hong-Linh Truong
Aalto University, Finland
linh.truong@aalto.fi



Ngoc Nhu Trang Nguyen
Daienso Lab, Vietnam
nhutrang.nguyen@daienso.com



07.09.2026



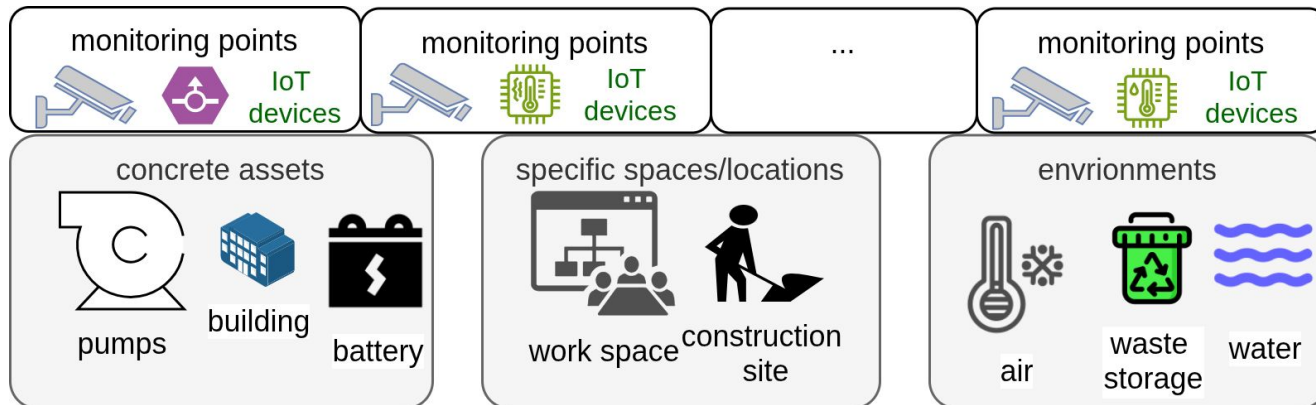
Content is available under
CC BY-SA 4.0 unless otherwise stated

Content

- Motivation
- The Analytics Feature Space (AFS) framework
- AFS-aware Platform services and implementation
- Experiments
 - operation management applications in the Intelligent Operation Center (IOC)
- Conclusion

Motivation scenarios: diverse edge AI/ML detection pipelines provide different services/results (1)

Factories, IOC, smart buildings, etc.: new developments and modernization



But the consumer mainly focuses on analytics of complex subjects

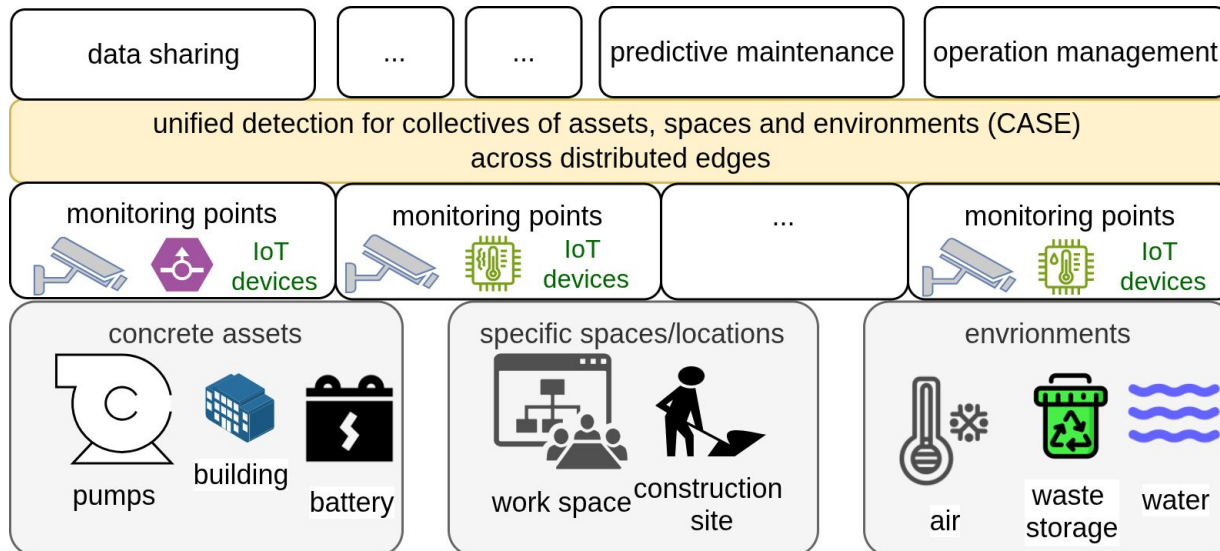
Different edge detection pipelines: object detection, anomaly detection, fault detection, etc.

Diversity: from multiple providers, with different lifecycle of engineering updates and deployment, various AI/ML models, and runtime environment changes

Dynamic changes: AI/ML models, configurations, services, and environments in *independent and uncoordinated* actions

Motivation scenarios: diverse edge AI/ML detection pipelines provide different services/results (2)

⇒ Unified detection management is needed for complex subjects at the edge



Empower interpretable and quality of data for downstream applications and data sharing for third-party data consumers

Manage metadata about different edge detection pipelines: in a unified way

Focus on complex analytics subjects: Collective of Assets, Spaces, and Environments (CASE**)**

Unified view of relevant software and data components

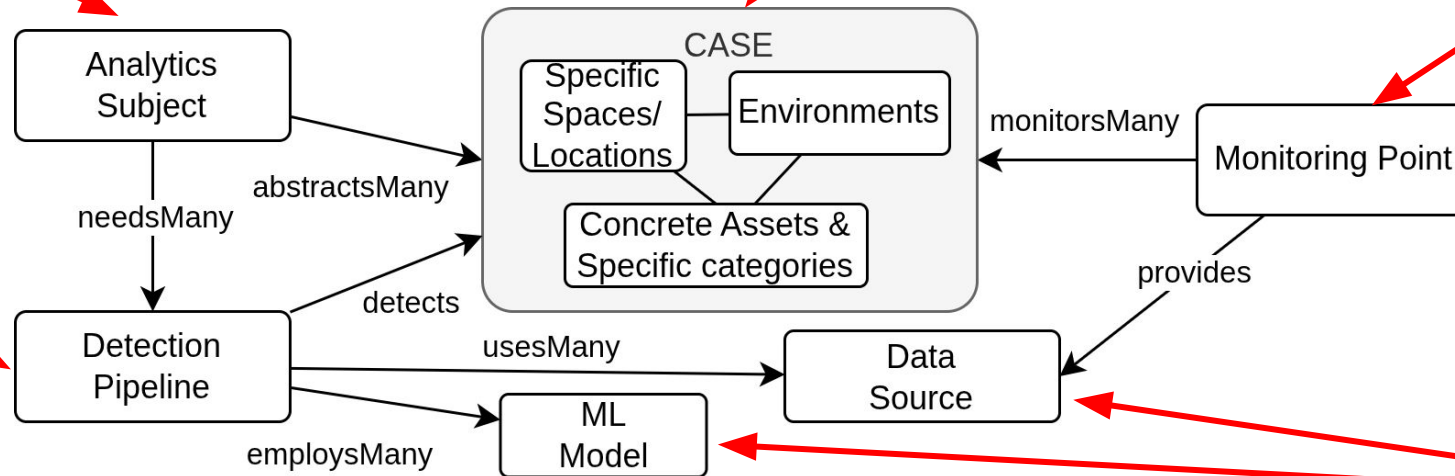
High-level abstraction indicates subject to be analyzed

Complex analytics and downstream applications for a CASE

Linked to different cameras/sensors/etc.

Various detection pipelines for dynamic, multiple types of detection

Different data sources and different AI/ML models



⇒ Besides the **detection results**: it is important to obtain various types of **metadata** and **changes in detection deployments** (shift left actions w.r.t. the diversity)

A!

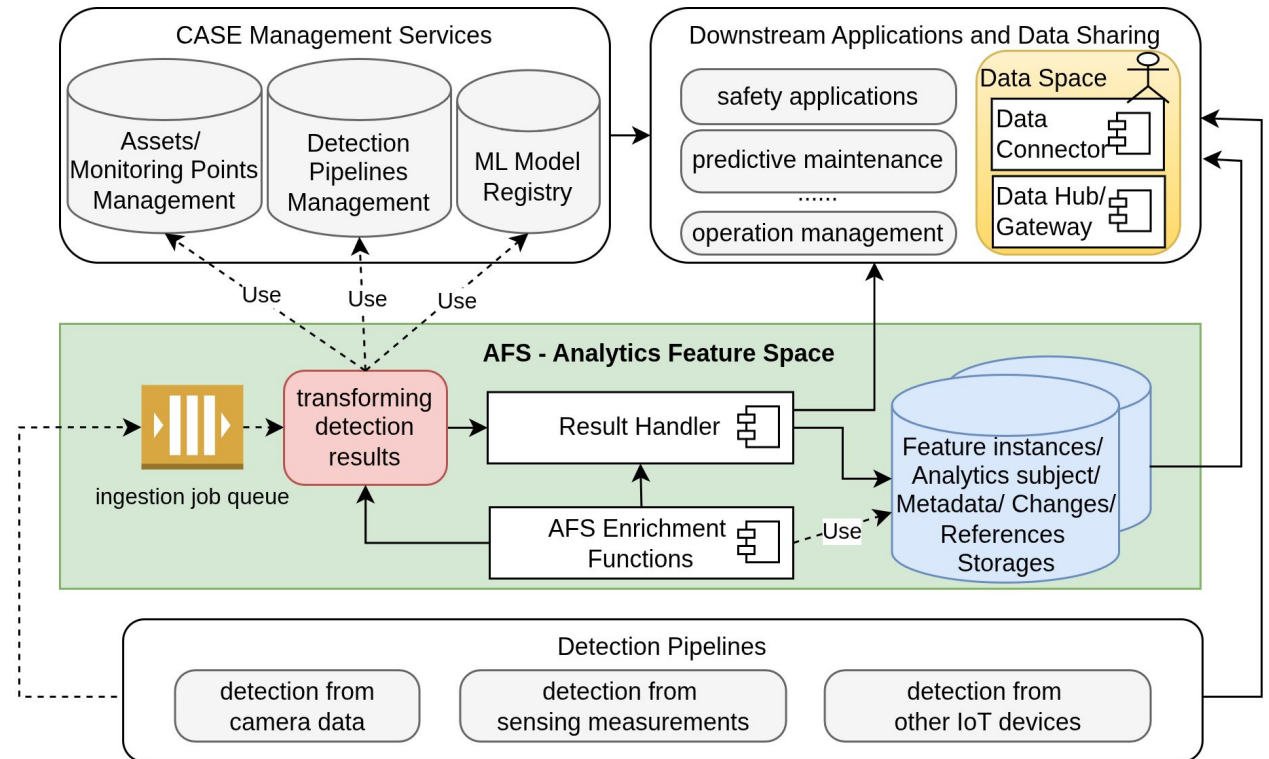
Research questions

- **RQ1: Which types of **metadata** and how to link them to the detection results and analytics for a CASE?**
 - common metadata reflecting different type of detections, AI/ML models, and assets in a CASE
 - metadata about AI/ML inferences/detection
- **RQ2: What types of **changes** need to be captured and exchanged?**
 - changes can happen in AI/ML-based detection and configurations and affect the downstream and explainability about the shared data
 - changes of constraints on detection results
 - changes in data sources
 - changes of AI/ML models

The Analytics Feature Space (AFS) framework

- Making several types of *metadata* associated with edge detection pipelines for a CASE *explicitly*
- **Capturing important changes** for downstream applications
- Developing various *functions* together with *messaging middleware* and other relevant components to support the **enrichment of detection results and metadata**

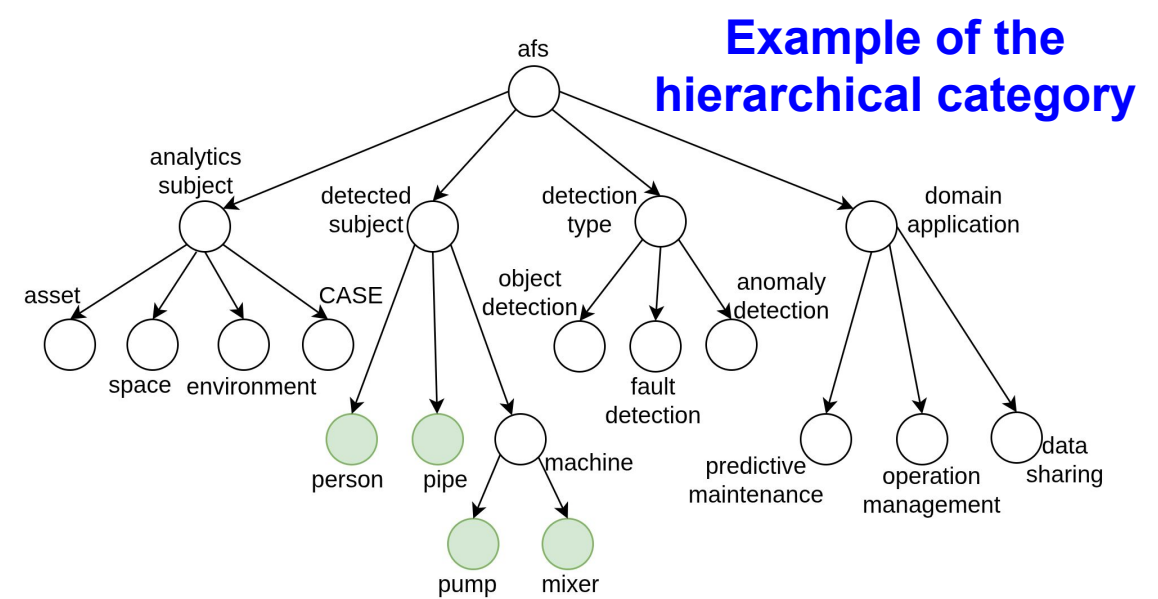
to address interoperability, integration, and explainability



Key entities of AFS have been published as open source:
<https://github.com/daienso/fansipan-afs>

Data Modeling in AFS

- Explicit ***Analytics Subject*** to represent a CASE for complex detection and analytics
- Explicit ***naming classes*** for detection results and ***purpose of the detection***
- Supports a ***hierarchical domain category*** and models the domain category hierarchy explicitly
- Explicit detection ***feature instances*** and ***metadata associated*** with the feature instances



```
class CommonDetectionResult(BaseModel):
    domainApplication: str = None
    detectionResult: Union[ObjectDetectionResult, dict]
    detectionSourceRefId: str = None
    additionalData: dict = None

class FeatureInstance(BaseModel):
    featureInstanceId: str
    timestamp: float
    detectionResult: List[CommonDetectionResult] = None
    analyticsSubject: AnalyticsSubject
    domainApplication: str = None
    detectionModel: DetectionModel = None
    detectionConfig: DetectionConfig = None
```

a schema for representing feature instances

AFS functions and messaging middleware

- **AFS functions**

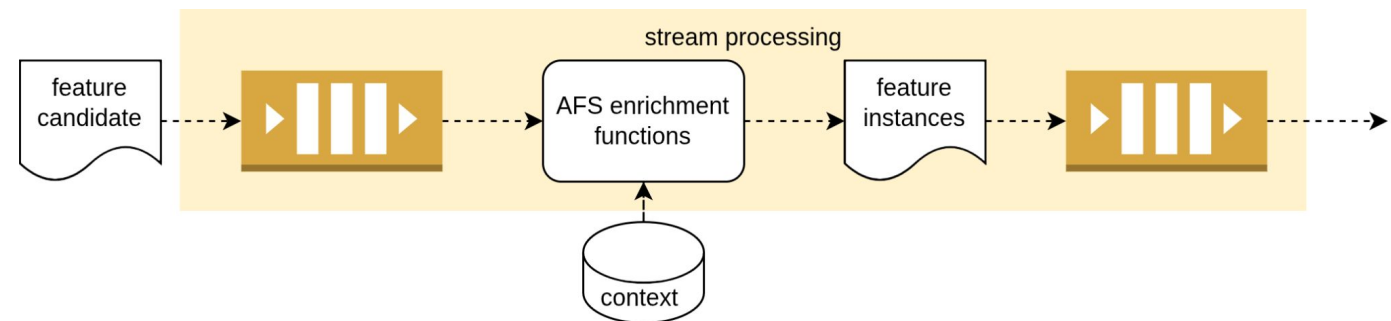
- Transformation/Enrichment functions
- Capturing runtime configuration changes
- Capturing data source changes

- **Messaging middleware for AFS (MAFS)**

- Context-aware result delivery
- Integration of functions for transforming/enriching results
- Delivery of additional data

```
class DataSource(BaseModel):  
    name: str=None  
    type: str=None  
    sourceRefId: str=None  
    properties: dict=None  
  
class DataSourceChange(BaseModel):  
    timestamp: float  
    analyticsSubject: AnalyticsSubject = None  
    changeType: ChangeTypeEnum = None  
    dataSources: List[DataSource]
```

Modeling changes in data sources



AFS enrichment functions deployed within messaging systems

Integration with pipeline development and Data Spaces

- **Metadata, detection results, and AFS models can be used differently**
 - the downstream applications use AFS to transform non-AFS results
 - existing pipelines can be modified to add metadata into detection results
 - the pipelines completely uses new schemas from AFS

```
class ModelComplianceDoc(BaseModel):  
    documentId: str  
    issuanceDate: Union[float, str]  
    expirationDate: Union[float, str]  
    issuer: str  
    holder: str  
    registryId: str  
    modelId: str  
    complianceScope:  
    dict[RegulationClassEnum, ComplianceResultClassEnum]  
    extra: dict=None
```

A schema for representing compliance information about the detection pipeline

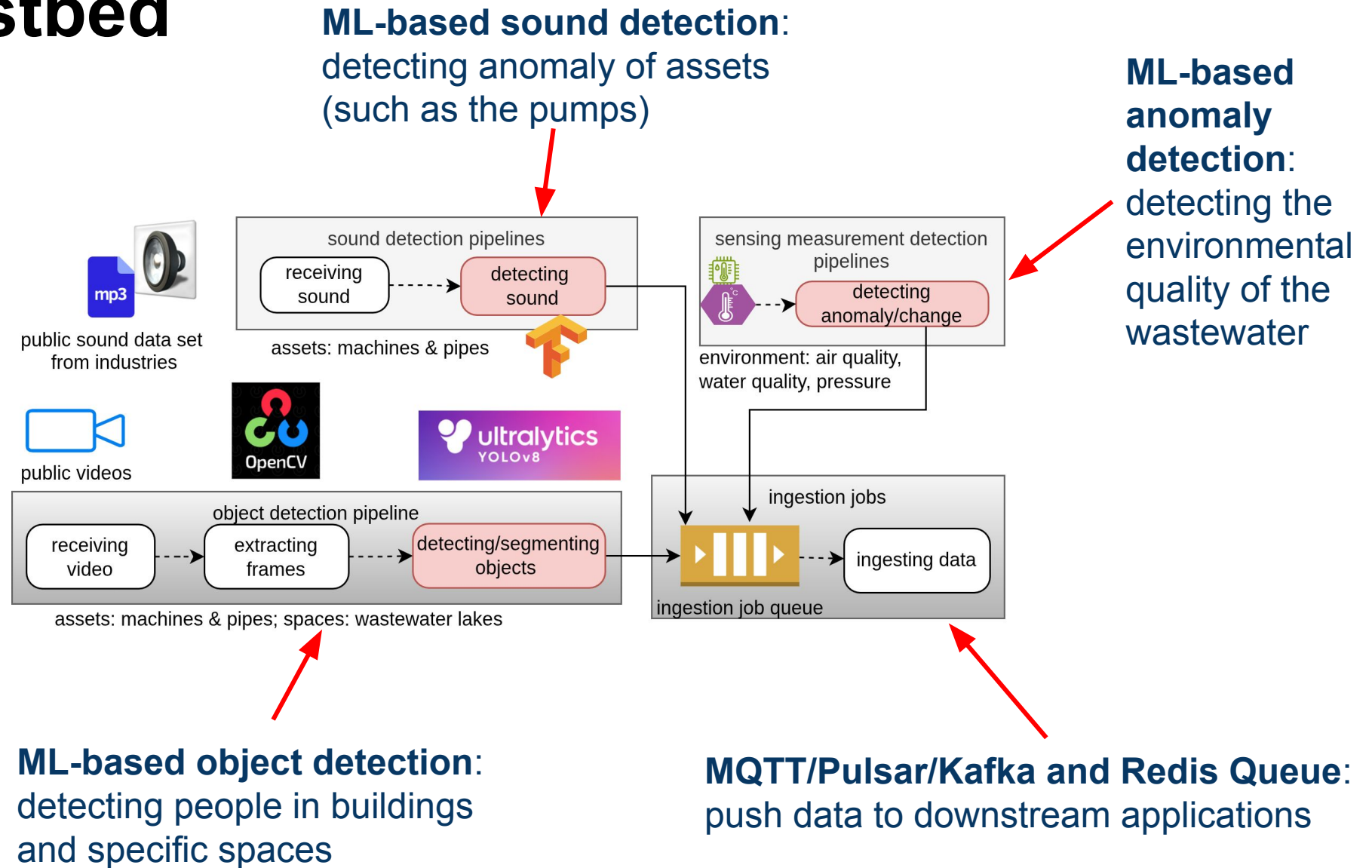
- **Integration with Data Spaces**
 - data connector for retrieving data
 - basic compliance information to enrich data product catalog and policies for data sharing

Experiments: Intelligent Operation Center (IOC) scenarios

- Downstream applications in our industrial collaborations (IOC services in the cloud for industrial zones):
 - Manage operations, environmental conditions, and safety in different wastewater treatment factories located in different places
- Upstream detection (edge detection):
 - Several edge cameras and IoT devices are deployed, managed, and operated autonomously with many ML-based detection pipelines, decoupled from the IOC services in the cloud
- Third-party data consumers:
 - During operation, many faults of assets can occur and be detected via alarms or fault/anomaly detection pipelines, thus requiring data sharing with the suppliers/vendors for maintenance

Emulation testbed

- Illustrative operation management application when an incident or problem happens in a specific analytics subject
- Data sharing for maintenance service
- Using open source data and AI/ML models



Experiments: using AFS data models and functions to support analytics and data sharing

```
"timestamp": 1759575183.752002,
"analyticsSubject": {
  "analyticsSubjectId":
    "as-case01wastewater"
},
"detectionModel": {
  "detectionModelName": "yolov8n.pt"
},
"config": {
  "result_constraints": {
    "person": {
      "operator": ">",
      "value": 2
    }
  }
}
```

Example of a change of result constraints

```
analytics_subject= "as-case01wastewater"
start_time = specified_time - interval/2
end_time = specified_time + interval/2
query = {
  "timestamp": {'$gte': start_time, '$lt':
    end_time},
  "analyticsSubject.analyticsSubjectId" :
    analytics_subject
}
results =
db_collection.find(query,projection=projection)
for result in results:
  #...
  #get image/evidence in the select result:
  temp_data_file=os.path.basename(selected_result
    ["ref_image"])
  #get the evidence image
  minio_utils.downloadtolocal(bucket,selected_res
    ult["ref_image"],temp_data_file)
```

Simple code illustrating the query of relevant evidences when a streaming detecting anomalies

```
"afs:complianceDoc": {
  "@type":
    "afs:ModelComplianceDoc" ,
  "documentId": "...",
  "issuanceDate":
    "2025-09-23T15:19:13.939524" ,
  "expirationDate":
    "2026-09-23T15:19:13.939536" ,
  "issuer": "IOC",
  "holder": "ioc-f01",
  "registryId":
    "reg:...",
  "modelId": "pump_id_04_6dB" ,
  "complianceScope": {
    "NIST AI RMF 1.0":
      "Conformant"
  },
}
```

Example of a custom compliance information

Conclusion

- AFS as a framework enabling the analytics of collectives of assets, spaces and environments (CASE)
 - abstract individual edge AI/ML-based detections and unified them for analytics of subjects associated with CASE
 - various schemas and techniques to link and enrich metadata for detection integration and interoperability
- Future work
 - prototype and integration of AFS into real-world systems of intelligent operation centers
 - detection data enrichment and data sharing

A!

—

Kiitos
aalto.fi